

Case Study #4:

Research Project Management Office (RMPO)

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about research projects, grants, donors, and the faculty, staff, and students involved.

OBJECTIVE

1. Maintain Data with CRUD method

Administrators can retrieve, review, and update information within the Research Project Management Office (RMPO) system by using standard CRUD (Create, Read, Update, Delete) operations to ensure accurate and up-to-date database management.

2. Analyse and report statistics for management

Administrators can utilize the data stored in the system to perform statistical analysis and generate reports. These insights can serve as reliable references to support decision-making and strategic management.

3. Tracking, planning and forecasting

Administrators can track the progress of each research project and leverage project data for planning purposes. The system also allows forecasting of project timelines, required resources, or budgets to improve efficiency and project outcomes.

4. Research performance of each faculty

Administrators can evaluate the research performance of each faculty or department, enabling the assessment of capabilities, achievements, and overall contributions to institutional research success.

Kookle University
Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

Keywords: Ontology, Plant Disease, Intelligent System

Project Category: ☐ Research and Development
☐ Training/Capacity Building
☐ Conference/Workshop

Research Areas of the Project: Semantic Web, Ontology, Knowledge Representation

Project Duration: 12 months

Contract Period: 1 Jan 2019 – 31 Dec 2019

Budget: 1,000,000 THB

Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:

1. Name: Mark Zuckerberg Email: mark@kmail.com
Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
Department ICT Department Area of Expertise: Knowledge Representation

Project Member:

1. Name: Tim Berners Lee Email: tim@kmail.com
Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: _____ Email: _____
Department _____ Area of Expertise: _____

Donor/Sponsor Name: Thailand Research Fund

Donor Type: ☐ Research Organization
☐ International Organization
☐ Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:

1. Name: Nina Billa Email: nina@kmail.com
Department: ICT
Responsibility: Programming
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
Department: Business Management
Responsibility: Project Coordinator
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Jan 2019 End date: 31 Dec 2019

Based on the form, the requirements for the Kookle University Research Project Management Office (RPMO) system are as follows:

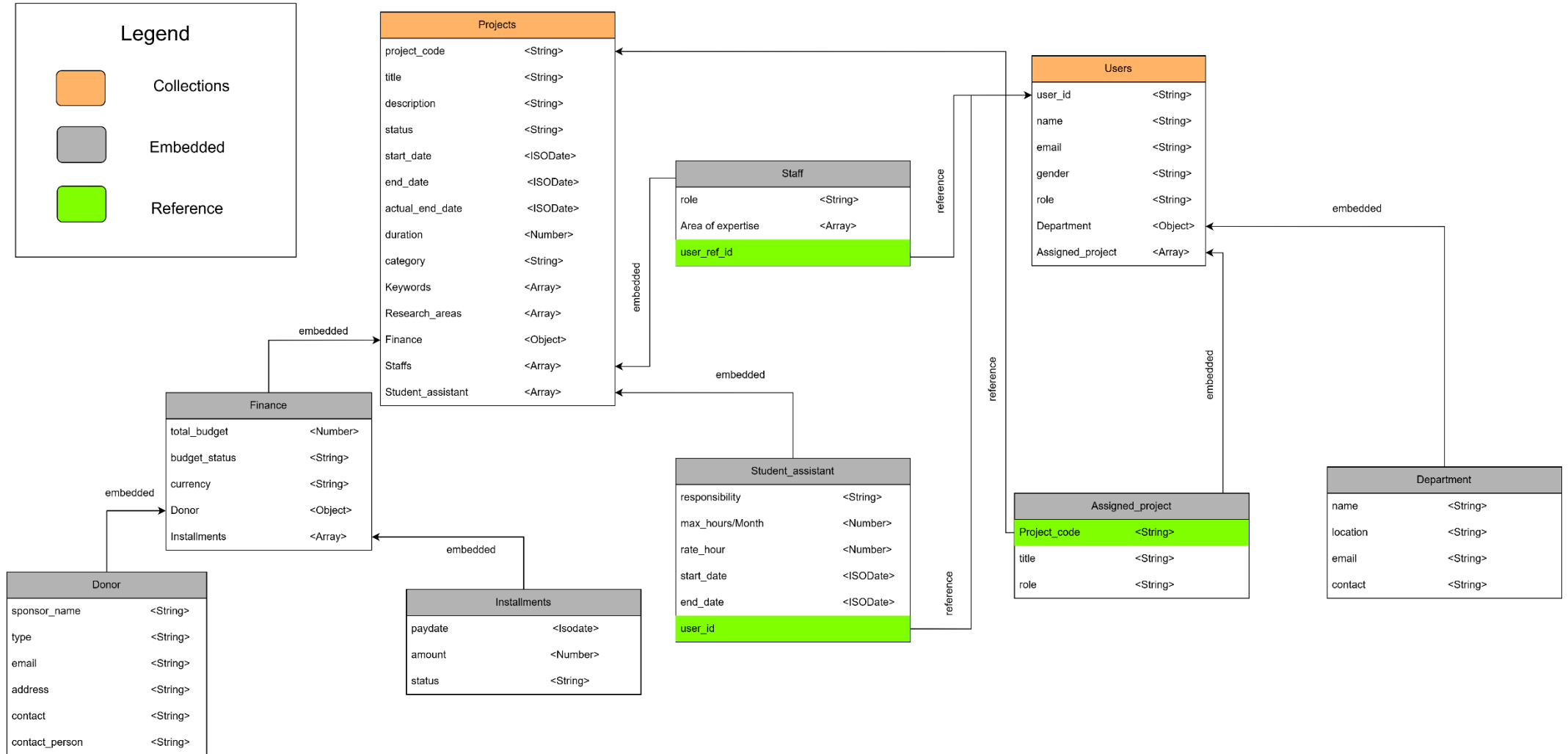
Data Maintenance: The system must maintain data about research projects, grants received, donors, and the faculty, staff, and students working on the projects.

Data Entry and Updates: It should support the entry and updating of project-related data after a project form is submitted.

Querying: The system needs to allow for querying data from project spreadsheets. This includes queries to coordinate with the project investigator.

Reporting: The system should be able to generate useful reports or statistics for university management. Examples of reports include total grants received for each active research area, statistics about sponsors/donors, and the research performance of each faculty/staff member.

Document- diagram



JSON SCHEMA

Users Collection

```
{
  "user_id": <Number>,
  "name": <String>,
  "email": <String>,
  "contact": <String>,
  "gender": <String>,
  "department": {
    "name": <String>,
    "location": <String>,
    "email": "<String>",
    "contact": <String>
  },
  "assigned_projects": [
    {
      "project_code": <String>,
      "title": <String>,
      "role": <String>
    }
  ]
}
```

Database – MongoDB Server

mongodb+srv://st125843:Ny47gEN86MZWFrAa@dmmcluster0.xqqaey.mongodb.net/

Projects Collection

```
{  
  "project_code": <String>,  
  "title": <String>,  
  "description": <String>,  
  "category": <String>,  
  "status": <String>,  
  "start_date": <ISODate>,  
  "end_date": <ISODate>,  
  "duration": <Number>,  
  "actual_end_date": <Date>,  
  "research_areas": <Array of String> ,  
  "keywords": <Array of String> ,  
  "staff": [ { "user_ref_id": <user_id>  
    "role": <String>,  
    "area_of_expertise": <Array of String> } ],  
  "student_assistant": [  
    { "user_ref_id": <user_id>,  
      "responsibility": <String>,  
      "max_working_hr_per_m": <Number>,  
      "rate_per_hr": <Number>,  
      "start_date": <ISODate>,  
      "end_date": <ISODate>
```

```
    "end_date": <ISODate> } ],  
  "financials": {  
    "total_budget": <Number>,  
    "currency": <String>  
    "budget_status": <String>,  
    "donor": {  
      "sponsor_name": <String>,  
      "type": <String>,  
      "email": <String>,  
      "address": <String>,  
      "contact_person": <String>,  
      "contact": <String>  
    },  
    "installments": [  
      {  
        "pay_date": <ISODate>,  
        "amount": <Number>,  
        "status": <String>  
      }  
    ]  
  }  
}
```

QUERY

1. Find all active projects in "Research and Development" category

NoSQL Command:

```
db.projects.find({category:"Research and  
Development",status:"Active"},{_id:0,project_code:1,title:1,category:1,status:1,duration:1,start_date:1,end_date:1});
```

```
{  
  project_code: 'AI-AGRI-2025',  
  title: 'AI Application for Precision Agriculture',  
  category: 'Research and Development',  
  status: 'Active',  
  start_date: 2025-01-15T00:00:00.000Z,  
  end_date: 2026-12-31T00:00:00.000Z,  
  duration: 24  
}  
{  
  project_code: 'SMART-CITY-2025',  
  title: 'Smart City IoT Integration and Analytics Platform',  
  category: 'Research and Development',  
  status: 'Active',  
  start_date: 2025-03-01T00:00:00.000Z,  
  end_date: 2027-02-28T00:00:00.000Z,  
  duration: 24  
}
```

2.List all projects funded by donors of type "Research Organization"

NoSQL Command:

```
db.projects.find ({"financials.donor.type":"Research Organization"},  
{_id:0,project_code:1,title:1,"financials.donor.sponsor_name": 1, "financials.donor.type" : 1 } );
```

```
{  
  project_code: 'AI-AGRI-2025',  
  title: 'AI Application for Precision Agriculture',  
  financials: {  
    donor: {  
      sponsor_name: 'National Research Council of Thailand',  
      type: 'Research Organization'  
    }  
  }  
}
```

3. Calculate total budget amount by research area for current year

NoSQL Command:

db.projects.aggregate

```
([{$match:{start_date:{$lte:ISODate("2025-12-31T00:00:00Z")},end_date:{$gte:ISODate("2025-01-01T00:00:00Z")}}},
{$unwind:"$research_areas"},
{$group:{_id:"$research_areas",total_budget: { $sum:"$financials.total_budget" }, project_count:{$sum:1}}},
{$sort:{total_budget:-1}}]);
```

```
{
  _id: 'Data Science',
  total_budget: 3000000,
  project_count: 1
}
{
  _id: 'Urban Analytics',
  total_budget: 3000000,
  project_count: 1
}
{
  _id: 'Internet of Things',
  total_budget: 3000000,
  project_count: 1
}
{
  _id: 'Artificial Intelligence',
  total_budget: 2500000,
  project_count: 1
}
{
  _id: 'Agriculture',
  total_budget: 2500000,
  project_count: 1
}
```

4. Show project details with donor information and installment schedule

NoSQL Command:

```
db.projects.find({}, {_id:0,project_code:1,title:1,category:1,status:1,"financials.donor":1,"financials.installments":1});
```

```
{
  project_code: 'AI-AGRI-2025',
  title: 'AI Application for Precision Agriculture',
  category: 'Research and Development',
  status: 'Active',
  financials: {
    donor: {
      sponsor_name: 'National Research Council of Thailand',
      type: 'Research Organization',
      email: 'contact@nrct.go.th',
      address: '196 Phahonyothin Rd, Bangkok',
      contact_person: 'Mr. Somchai Rakdee',
      contact: '02-561-2445'
    },
    installments: [
      {
        pay_date: 2025-02-01T00:00:00.000Z,
        amount: 1250000,
        status: 'Received'
      },
      {
        pay_date: 2026-02-01T00:00:00.000Z,
        amount: 1250000,
        status: 'Pending'
      }
    ]
  }
}
```

```
{
  project_code: 'ROBO-MED-2023',
  title: 'Robotic Arm for Minimally Invasive Surgery',
  category: 'Conference/Workshop',
  status: 'Completed',
  financials: {
    donor: {
      sponsor_name: 'Ministry of Public Health',
      type: 'Private Company',
      email: 'contact@moph.go.th',
      address: '88/20 Tiwanon Rd, Nonthaburi',
      contact_person: 'Dr. Supoj Meesuk',
      contact: '02-590-1000'
    },
    installments: [
      {
        pay_date: 2023-02-15T00:00:00.000Z,
        amount: 1750000,
        status: 'Received'
      },
      {
        pay_date: 2024-02-15T00:00:00.000Z,
        amount: 1750000,
        status: 'Received'
      }
    ]
  }
}
```

5. Identify projects ending within next 3 months

NoSQL Command:

```
db.projects.find ( {end_date: { $gte: ISODate("2025-10-13T00:00:00Z"), $lte: ISODate("2026-01-13T00:00:00Z")}},  
{"project_code": 1, "title": 1, "end_date": 1, "status": 1});
```

```
{  
  _id: ObjectId('68eccfb8aaf061da3f0a00b2'),  
  project_code: 'URBAN-TRANS-2024',  
  title: 'Sustainable Urban Transportation Planning',  
  status: 'Active',  
  end_date: 2025-12-20T00:00:00.000Z  
}
```

6. Calculate average project duration by category

NoSQL Command:

```
db.projects.aggregate([  
  { $group: { _id: "$category", average_duration_in_months: { $avg: "$duration" }}}  
])
```

```
< {  
  _id: 'Training/Capacity Building',  
  average_duration_in_months: 24  
}  
{  
  _id: 'Research and Development',  
  average_duration_in_months: 24  
}  
{  
  _id: 'Conference/Workshop',  
  average_duration_in_months: 24  
}  
{  
  _id: 'Fundamental Research',  
  average_duration_in_months: 24  
}
```

7. List all international organization donors and their contact details

NoSQL Command:

```
db.projects.aggregate([
  { $match: { "financials.donor.type": "International Organization" } },
  { $project: { _id: 0, sponsor_name: "$financials.donor.sponsor_name",
contact_person: "$financials.donor.contact_person", email: "$financials.donor.email",
contact_number: "$financials.donor.contact", address: "$financials.donor.address" }}
])
```

```
< {
  sponsor_name: 'PTT Global Chemical',
  contact_person: 'Ms. Areeya Chansuk',
  email: 'info@pttggroup.com',
  contact_number: '02-265-8400',
  address: '555/1 Energy Complex, Bangkok'
}
{
  sponsor_name: 'Thai Health Promotion Foundation',
  contact_person: 'Mr. Eakachai Lertpong',
  email: 'info@thaihealth.or.th',
  contact_number: '02-343-1500',
  address: '99/8 Sathorn Nuea Rd, Bangkok'
}
```

8. Calculate total funding received from each donor type by year

NoSQL Command:

```
db.projects.aggregate([
  { $unwind: "$financials.installments" },
  { $match: { "financials.installments.status": "Received" } },
  {$group: { _id: { donor_type: "$financials.donor.type", year: { $year: "$financials.installments.pay_date" } },
    total_funding: { $sum: "$financials.installments.amount" } }},
  {$project: { _id: 0, donor_type: "$_id.donor_type", year: "$_id.year", total_funding: "$total_funding" } },
  {$sort: { year: 1, donor_type: 1 } }
])
```

```
< {
  donor_type: 'International Organization',
  year: 2022,
  total_funding: 1100000
}
{
  donor_type: 'International Organization',
  year: 2023,
  total_funding: 1100000
}
{
  donor_type: 'Private Company',
  year: 2023,
  total_funding: 2500000
}
{
  donor_type: 'Private Company',
  year: 2024,
  total_funding: 3750000
}
```

9. Identify top 5 research areas by total funding received

NoSQL Command:

```
db.projects.aggregate([
  {$unwind: "$research_areas" },
  {$group: { _id: "$research_areas", total_funding: { $sum: "$financials.total_budget" } }},
  {$sort: { total_funding: -1 }},
  {$limit: 5},
  { $project: { _id: 0, research_area: "$_id", total_funding: "$total_funding" } }
])
```

```
< {
  research_area: 'AI',
  total_funding: 4000000
}
{
  research_area: 'Energy Systems',
  total_funding: 4000000
}
{
  research_area: 'Medical Technology',
  total_funding: 3500000
}
{
  research_area: 'Robotics',
  total_funding: 3500000
}
{
  research_area: 'Agriculture',
  total_funding: 2500000
}
```

10. Show project timeline and payment schedule

NoSQL Command:

```
db.projects.aggregate([
  { $match: { status: "Active" } },
  { $lookup: { from: "users", localField: "staff.user_ref_id", foreignField: "user_id", as: "staff_details" } },
  { $project: { _id: 0, project_code: 1, title: 1, staff_members: "$staff_details.name" } })
```

```
< {
  project_code: 'AI-AGRI-2025',
  title: 'AI Application for Precision Agriculture',
  staff_members: [
    'Dr. Wipa Saelim',
    'Dr. Somchai Makdee',
    'Dr. Malee Rukdee',
    'Somsak Decha'
  ]
}
{
  project_code: 'GRID-SOLAR-2024',
  title: 'Smart Grid Integration for Solar Power',
  staff_members: [
    'Dr. Wipa Saelim',
    'Apiwat Saelim',
    'Sunisa Srichai'
  ]
}
```

11. List projects with their completion percentage based on installments received

NoSQL Command:

```
db.projects.aggregate([ {$unwind: "$financials.installments"},
{$match: {"financials.installments.status": "Received"} },
{ $group: { _id: "$_id", project_title: { $first: "$title" }, total_budget: { $first: "$financials.total_budget" },
amount_received: { $sum: "$financials.installments.amount" } } },
{ $project: { _id: 0, project_title: 1, total_budget: 1, amount_received: 1,
completion_percentage: { $cond: { if: { $gt: ["$total_budget", 0] },then: { $multiply: [ { $divide: ["$amount_received",
"$total_budget" ] }, 100 ]},else: 0 } } } } ]})
```

```
>_MONGOSH
< {
  project_title: 'AI Application for Precision Agriculture',
  total_budget: 2500000,
  amount_received: 1250000,
  completion_percentage: 50
}
{
  project_title: 'Smart Grid Integration for Solar Power',
  total_budget: 4000000,
  amount_received: 2000000,
  completion_percentage: 50
}
{
  project_title: 'Robotic Arm for Minimally Invasive Surgery',
  total_budget: 3500000,
  amount_received: 3500000,
  completion_percentage: 100
}
{
  project_title: 'Sustainable Urban Transportation Planning',
  total_budget: 2500000,
  amount_received: 2500000,
```

12. Update project status when contract end date is reached

NoSQL Command:

```
db.projects.updateMany(  
  {"end_date": { $lte: new Date() }, "status": { $ne: "Completed" } },  
  {$set: { "status": "Completed" } });
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
{  
  _id: ObjectId('671e8a0e1b2c3d4e5f6a7b82'),  
  project_code: 'ROBO-MED-2023',  
  title: 'Robotic Arm for Minimally Invasive Surgery',  
  description: 'Development of a high-precision robotic arm for surgical procedures.',  
  category: 'Conference/Workshop',  
  status: 'Completed',  
  start_date: 2023-01-20T00:00:00.000Z,  
  end_date: 2024-12-20T00:00:00.000Z,  
  duration: 24,  
  actual_end_date: 2024-11-30T00:00:00.000Z,  
  research_areas: [  
    'Robotics',  
    'Medical Technology'  
  ],  
  keywords: [  
    'surgical robot',  
    'haptics'  
  ],  
}
```

13. Insert new installment record for approved project payments

NoSQL Command:

```
db.projects.updateOne({"project_code": "NANO-COAT-2023","financials.budget_status": "Approved"},  
  {$push: {"financials.installments": {"pay_date": ISODate("2025-11-15T00:00:00Z"),"amount": 750000.00,  
    "status": "Pending"}}});
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
{  
  financials: {  
    installments: [  
      {  
        pay_date: 2023-09-15T00:00:00.000Z,  
        amount: 750000,  
        status: 'Received'  
      },  
      {  
        pay_date: 2025-11-15T00:00:00.000Z,  
        amount: 750000,  
        status: 'Pending'  
      }  
    ]  
  }  
}
```

14. Delete project code = AI-AGRI-2025 and associated records

NoSQL Command:

```
db.projects.deleteOne({ "project_code": "AI-AGRI-2025" });
```

```
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

Result

```
> db.projects.findOne({ "project_code": "AI-AGRI-2025" });  
< null
```

15. Calculate monthly burn rate for active projects

NoSQL Command:

```
db.projects.aggregate([
  { $match: { "status": "Active" } },
  { $project: { _id: 0, project_code: 1, title: 1, monthly_burn_rate: { $cond: { if: { $gt: ["$duration", 0] },
    then: { $divide: ["$financials.total_budget", "$duration"] },
    else: 0 } } } } }];
```

```
{
  project_code: 'URBAN-TRANS-2024',
  title: 'Sustainable Urban Transportation Planning',
  monthly_burn_rate: 104166.66666666667
}
{
  project_code: 'GRID-SOLAR-2024',
  title: 'Smart Grid Integration for Solar Power',
  monthly_burn_rate: 166666.66666666666
}
{
  project_code: 'AI-AGRI-2025',
  title: 'AI Application for Precision Agriculture',
  monthly_burn_rate: 104166.66666666667
}
```

16. Track project keyword trends over the past 5 years

NoSQL Command:

```
db.projects.aggregate([
  { $match: { "start_date": { $gte: new Date("2020-10-13") } } },
  { $unwind: "$keywords" },
  { $group: { _id: { year: { $year: "$start_date" }, keyword: "$keywords" }, count: { $sum: 1 } } },
  { $group: { _id: "$_id.year", keyword_trends: { $push: { keyword: "$_id.keyword", count: "$count" } } } },
  { $sort: { "_id": 1 } },
  { $project: { _id: 0, year: "$_id", trends: "$keyword_trends" } } ])
```

```
< {
  year: 2022,
  trends: [
    {
      keyword: 'big data',
      count: 1
    },
    {
      keyword: 'predictive analytics',
      count: 1
    }
  ]
}
{
  year: 2023,
  trends: [
    {
      keyword: 'surgical robot',
      count: 1
    },
    {
      keyword: 'nanoparticles',
      count: 1
    },
    {
      keyword: 'corrosion',
      count: 1
    },
    {
      keyword: 'haptics',
      count: 1
    }
  ]
}
```

17. For all ongoing projects, show the total number of installments, the number of incomplete installments, and the number of complete installments.

NoSQL Command:

```
{ $match: { "status": { $in: ["Active", "On Hold"] } } },
{ $project: { _id: 0, project_code: 1, title: 1, total_installments: { $size: "$financials.installments"
}, complete_installments: {
$size: { $filter: { input: "$financials.installments", as: "item", cond: { $eq: [ "$$item.status", "Received" ] } } } },
incomplete_installments: { $size: { $filter: { input: "$financials.installments", as: "item", cond: { $eq: [ "$$item.status",
"Pending" ] } } } } } } }
```

```
< {
  project_code: 'AI-AGRI-2025',
  title: 'AI Application for Precision Agriculture',
  total_installments: 2,
  complete_installments: 1,
  incomplete_installments: 1
}
{
  project_code: 'GRID-SOLAR-2024',
  title: 'Smart Grid Integration for Solar Power',
  total_installments: 2,
  complete_installments: 1,
  incomplete_installments: 1
}
{
  project_code: 'NANO-COAT-2023',
  title: 'Nanotechnology-based Anti-Corrosion Coating',
  total_installments: 1,
  complete_installments: 1,
  incomplete_installments: 0
}
{
  project_code: 'URBAN-TRANS-2024',
  title: 'Sustainable Urban Transportation Planning',
  total_installments: 2,
  complete_installments: 2,
  incomplete_installments: 0
}
```

18. Identify projects that have passed their contract end date but are still 'On going'.

NoSQL Command:

```
db.projects.find({  
  
  "status": { $in: ["Active", "On Hold"] },  
  
  "end_date": { $lt: new Date("2025-10-13") }  
  
})
```

```
< {  
  _id: ObjectId('68eccc81570f18faeea6f7e2'),  
  project_code: 'NANO-COAT-2023',  
  title: 'Nanotechnology-based Anti-Corrosion Coating',  
  description: 'Creating a new coating material using nanoparticles to prevent metal corrosion.',  
  category: 'Conference/Workshop',  
  status: 'On Hold',  
  start_date: 2023-09-01T00:00:00.000Z,  
  end_date: 2025-08-31T00:00:00.000Z,  
  duration: 24,
```

19. For all ongoing projects, calculate the total estimated budget to be spent in each year (based on the contract end date)

NoSQL Command:

```
db.projects.aggregate([
  { $match: { "status": { $in: ["Active", "On Hold"] } } },
  { $addFields: { "daily_budget": { $divide: [ "$financials.total_budget", { $add: [{ $dateDiff: { startDate: "$start_date", endDate:
"$end_date", unit: "day" } }, 1] } ] } } },
  { $addFields: { "active_years": { $range: [ { $year: "$start_date" }, { $add: [{ $year: "$end_date" }, 1] } ] } } },
  { $unwind: "$active_years" },
  { $addFields: {
    "year_start": { $dateFromParts: { 'year': "$active_years", 'month': 1, 'day': 1 } },
    "year_end": { $dateFromParts: { 'year': "$active_years", 'month': 12, 'day': 31 } } } },
  { $addFields: {
    "effective_start_date": { $max: ["$start_date", "$year_start"] },
    "effective_end_date": { $min: ["$end_date", "$year_end"] }
  } },
  { $addFields: { "days_in_year": { $add: [{ $dateDiff: { startDate: "$effective_start_date", endDate: "$effective_end_date", unit:
"day" } }, 1] } } },
  { $group: { _id: "$active_years", total_estimated_budget: { $sum: { $multiply: ["$daily_budget", "$days_in_year"] } } } },
  { $project: { _id: 0, year: "$_id", total_estimated_budget: { $round: ["$total_estimated_budget", 2] } } },
  { $sort: { "year": 1 } }
])
```

```
< {  
  year: 2023,  
  total_estimated_budget: 250342  
}  
{  
  year: 2024,  
  total_estimated_budget: 3665256.15  
}  
{  
  year: 2025,  
  total_estimated_budget: 4986672.84  
}  
{  
  year: 2026,  
  total_estimated_budget: 1597729.01  
}
```

20. For each ongoing project, show the project lead (Principal Investigator), the total number of staff, and the project budget, to help in planning staff for new projects with similar budgets.

NoSQL Command:

```
db.projects.aggregate([
  { $match: { "status": { $in: ["Active", "On Hold"]} } },
  { $lookup: { from: "users", localField: "staff.user_ref_id", foreignField: "user_id", as: "staff_details" } },
  { $project: { _id: 0, project_code: 1, title: 1, project_lead: { $let: { vars: { pi_user_id:
{ $arrayElemAt: [{ $filter: { input: "$staff", as: "member", cond: { $eq: [ "$$member.role", "Principal Investigator" ] } } },
0] } },
in: { $arrayElemAt: [ { $filter: { input: "$staff_details", as: "staff_member", cond: { $eq: [ "$$staff_member.user_id",
"$pi_user_id.user_ref_id" ] } } }, 0] } },
total_staff: { $size: "$staff" }, project_budget: "$financials.total_budget" } },
{ $project: { project_code: 1, title: 1, project_lead: "$project_lead.name", total_staff: 1, project_budget: 1 } }
])
```

```
< {  
  project_code: 'AI-AGRI-2025',  
  title: 'AI Application for Precision Agriculture',  
  total_staff: 4,  
  project_budget: 2500000,  
  project_lead: 'Dr. Wipa Saelim'  
}  
{  
  project_code: 'GRID-SOLAR-2024',  
  title: 'Smart Grid Integration for Solar Power',  
  total_staff: 3,  
  project_budget: 4000000,  
  project_lead: 'Dr. Wipa Saelim'  
}  
{  
  project_code: 'NANO-COAT-2023',  
  title: 'Nanotechnology-based Anti-Corrosion Coating',  
  total_staff: 1,  
  project_budget: 1500000,  
  project_lead: 'Dr. Chan Chai'  
}  
{  
  project_code: 'URBAN-TRANS-2024',  
  title: 'Sustainable Urban Transportation Planning',  
  total_staff: 2,  
  project_budget: 2500000,  
  project_lead: 'Dr. Malee Rukdee'  
}
```

21. Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

NoSQL Command:

```
db.projects.aggregate([{$match:{status:{$in:["Active","On Hold"]}}},  
  
{$unwind:"$staff"},  
  
{$unwind:"$staff.area_of_expertise"},  
  
{$group:{_id:"$staff.area_of_expertise",total_staff:{$sum:1}}},  
  
{$sort:{total_staff:1}},{$limit:5},  
  
{$project:{_id:0,area_of_expertise:"$_id",total_staff:1}}]);
```

```
{  
  total_staff: 1,  
  area_of_expertise: 'Data Analytics'  
}  
  
{  
  total_staff: 1,  
  area_of_expertise: 'Data Science'  
}  
  
{  
  total_staff: 1,  
  area_of_expertise: 'AI'  
}  
  
{  
  total_staff: 1,  
  area_of_expertise: 'GIS'  
}  
  
{  
  total_staff: 1,  
  area_of_expertise: 'Urban Computing'  
}
```

22. Identify available staff members (not currently part of an ongoing project) who can be assigned to new projects.

NoSQL Command:

db.users.aggregate

```
([{$lookup:{from:"projects",localField:"user_id",foreignField:"staff.user_ref_id",as:"assigned_projects"}},
{$addFields:{active_projects:{$filter:{input:"$assigned_projects",as:"proj",cond:{$in:["$$proj.status",["Active","On Hold"]]]}}}},
{$match:{$expr:{$eq:[$size:"$active_projects"],0}}}},
{$project:{_id:0,user_id:1,name:1,email:1,department:1}}])
```

```
{
  user_id: 3,
  name: 'Natcha Petchdee',
  email: 'natcha.p@student.university.ac.th',
  department: {
    name: 'Faculty of Engineering',
    location: 'Building 4, Room 201',
    email: 'eng@university.ac.th',
    contact: '02-345-6789'
  }
}
{
  user_id: 4,
  name: 'Anan Jaidee',
  email: 'anan.j@student.university.ac.th',
  department: {
    name: 'Faculty of Science',
    location: 'Building 2, Room 110',
    email: 'sci@university.ac.th',
    contact: '02-345-6788'
  }
}
{
  user_id: 9,
  name: 'Pornthip Jaidee',
  email: 'pornthip.j@university.ac.th',
  department: {
    name: 'Faculty of Arts'
  }
}
```